SYSTEM FOR A DATA CENTER ARCHITECTURE WITH FUNCTIONALLY SEGREGATED RESOURCE POOLS

Reinaldo Javier Aguiar Marcano

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Title: SYSTEM FOR A DATA CENTER ARCHITECTURE WITH FUNCTIONALLY SEGREGATED RESOURCE POOLS

Inventor: Reinaldo Javier Aguiar Marcano, a citizen of the United States, residing at 2302 Britton Ridge Drive, Katy, TX 77494.

Abstract

A system for a data center architecture comprising functionally segregated zones, wherein each zone is dedicated to housing a specific type of computer component and is provided with infrastructure optimized for that component type. The system includes a first zone with high-density cooling for housing processors, a second zone with power-efficient infrastructure for housing memory modules, and a third zone optimized for physical density and robotic access for housing storage devices. An ultra-high-bandwidth data fabric interconnects all zones, allowing components from different zones to be logically composed into unified computing systems. This architecture maximizes operational efficiency by tailoring the physical infrastructure to the specific needs of each component type.

Background of the Invention

Traditional data centers are filled with racks of homogenous, monolithic servers. Each server is a self-contained computer, and the infrastructure for each rack (power, cooling, networking) must be designed to handle the server's peak load, which is typically dictated by its most demanding component (the CPU). This leads to widespread inefficiency, as the power and cooling infrastructure needed for the CPUs is over-provisioned for the less demanding memory and storage components within the same server chassis.

Summary of the Invention

The present invention is a data center architecture designed around the principle of functional segregation, enabled by a disaggregated computing model. The architecture abandons the monolithic server rack in favor of specialized zones or rows, each dedicated to a specific class of component.

The system comprises multiple distinct physical zones. A "Compute Zone" would consist of racks densely packed with processor chassis, serviced by high-performance liquid cooling systems. A "Memory Zone" would consist of racks of memory module chassis, designed with a focus on power efficiency and minimal cooling. A "Storage Zone" would consist of ultra-dense racks of storage devices, designed to allow for easy physical access by robotic systems. These and other specialized zones are all interconnected by a building-scale, ultra-high-bandwidth optical data fabric. This allows a control plane to dynamically compose a "computer" by linking a processor from the Compute Zone with memory from the Memory Zone and storage from the Storage Zone, creating a more efficient, scalable, and flexible data center.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a top-down schematic of the data center architecture, showing the functionally segregated resource zones interconnected by a central data fabric.

Detailed Description of the Invention

Referring to **FIG. 1**, a schematic of the data center architecture is shown. The data center floor is divided into functionally segregated zones. A first zone, the Compute Zone (101), contains racks of processors and is equipped with specialized, high-density cooling infrastructure. A second zone, the Memory Zone (102), contains racks of memory modules and is optimized for power efficiency. A third zone, the Storage Zone (103), contains ultra-dense racks of storage devices and may be serviced by robotic systems. All zones are interconnected by a central, high-bandwidth optical data fabric (104), which is managed by a data center control plane. This fabric allows any component in any zone to be logically connected to any other component in any other zone to compose logical computer systems on demand.

Claims

What is claimed is:

1. A data center system, comprising:

    - a first physical zone dedicated to housing a plurality of processors, said first zone having a first type of infrastructure optimized for processors;
    - a second physical zone, separate from the first, dedicated to housing a plurality of memory modules, said second zone having a second type of infrastructure optimized for memory modules; and
    - a data fabric interconnecting the first physical zone and the second physical zone.

2. The system of claim 1, wherein the first type of infrastructure is a high-density cooling system.

3. The system of claim 1, wherein the second type of infrastructure is a power-efficient distribution system.

4. The system of claim 1, further comprising a third physical zone dedicated to housing a plurality of storage devices, said third zone having a third type of infrastructure optimized for physical density and robotic access.

5. The system of claim 1, wherein the data fabric is an optical data fabric capable of dynamically connecting a processor in the first zone with a memory module in the second zone to form part of a logical computer.

6. The system of claim 5, wherein the optical data fabric uses interconnects as described in U.S. Patent Application No. 19/559,705.

Drawings

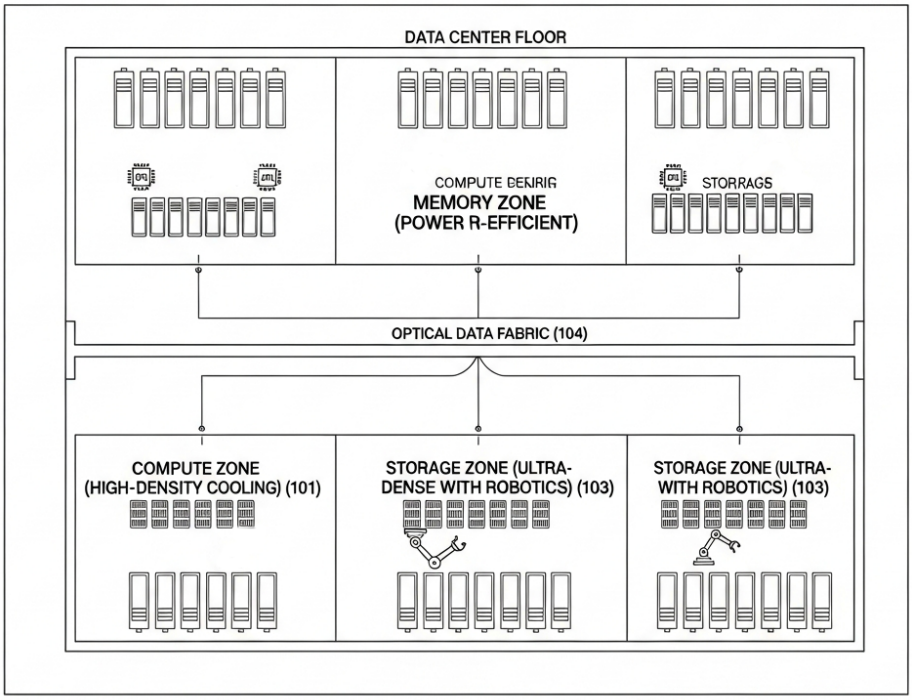


Figure 1: Top-down schematic of the data center architecture, showing the functionally segregated resource zones interconnected by a central data fabric.